

Bank Failures and the Effect on Geographically Proximate Institutions

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Abstract

This paper examines the localized contagion effects of U.S. bank failures on neighboring bank's stock returns from 2015 to 2025. Using 37 commercial bank failures and an event study framework based on the Fama-French Three Factor model, we find that proximate banks experience significant negative abnormal returns following failure announcements. The effects are strongest immediately post announcement and are robust across specifications. Our results highlight persistent regional contagion risks in banking markets, with implications for investors and regulators.

I. Introduction

The collapse of financial institutions has historically been a critical concern for economies, policymakers, and investors alike. Bank failures not only disrupt the immediate stakeholders but also send ripples through the broader financial system, sometimes triggering widespread panic and systemic risk. This has been suggested in other papers as the 'contagion' effect of bank failures. This is when the collapse of banks, especially a bigger bank, may cause the loss of public confidence in the banking system as a whole and is thus likely to set off runs on other banks (Aharony & Swary, 1983).

Understanding the causes and consequences of these failures is thus essential to ensuring financial stability. This paper explores a key dimension of financial contagion: the localized effects of bank

failures on the stock performance of other geographically proximate banks. While it is well understood that financial systems are interconnected, the extent to which the collapse of one institution triggers adverse outcomes for nearby banks remains an open question. We focus specifically on whether bank failures produce statistically significant negative impacts on the stock returns of nearby, publicly traded banks. Using a dataset of 37 U.S commercial banks failures between 2015 and 2025 and applying an event study methodology anchored in the Fama-French Three Factor Model, we measure abnormal returns around the time of each failure. Our analysis examines stock price movements across a 21-day event window, spanning 10 days before and after each bank's failure announcement.

By uncovering the localized market consequences of bank failures, this study aims to contribute important new insights into the dynamics of financial contagion and the broader risks facing modern banking systems. In an era where regional shocks can rapidly escalate into national crises, understanding these ripple effects is vital for investors, regulators, and scholars alike.

II. Literature Review

Rose, Peter S and Scott, William L (1980) researched eleven different commercial banks of the USA. The research paper carried out a detailed analysis to analyze various reasons behind bank failures by evaluating banks in various metrics like asset turnover, after-tax margin and financial leverage and the internal managements of a company. Their research paper concluded that the major reason behind the failure of large banks was various endogenous variables like quality and timeframe of management control.

After carrying out a detailed study of failed banks, banks facing problems, and healthy banks, the office of the comptroller of currency (1988) found that the main factor leading to a bank failure

was internal management. Contrary to the hypothesis that the major factor for bank failure was economic conditions, this research paper strongly advocates the importance of internal management, reforms and policies, decisions of board of directors, and controls to prevent banks from failure.

Alexandre Momparler, Pedro Carmona, and Francisco Climent (2020) tried to identify various factors that can cause bank failure in the USA by conducting empirical analysis using Fuzzy Set Qualitative Comparative Analysis (fsQCA). The research paper used various financial models like gradient boosting, neural networks, and various machine learning approaches that can be used to predict and correctly model the reasons for bank failure. They concluded that financial ratios are the major factor that indicates bank failure. The bank having a high share of bad loans with a low level of risk coverage showed a very high chance of failure.

M Brown, ST Trautmann, R Vlahu (2017) carried out an experimental coordination game to understand the phenomenon of a panicked run situation faced by a depositor of one bank and its impact on deposits of another bank. They concluded that the withdrawal of deposits from one bank is more likely to result in withdrawal of deposits from other banks.

Another research paper by Calomiris, Jaremski, and Wheelock (2019) carried out research to understand the contagion and interbank relationship during a crisis period, specifically during the great depression. Their study examined the impact of different contractual obligations that impacted the liquidity risk and led to a bank's failure. They found that the failure of one bank increases the likelihood of another bank failing, resulting from the shared distress that is transmitted from the failed bank.

Doyle (2020) used a modern neural network approach and a standard regression to explore the relationship between interbank contagion and default events. By using both methods, they concluded and verified that interbank contagion has a significant power to predict default events. However, they also showed that the use of a network contagion variable could significantly improve the model and better predict the reasons for bank failure.

Several past researchers have carried out in-depth studies on bank failures and their impact. Most of the earlier research has just explored the reason behind a bank's failure. Others have explored the intensity and impact of such bank failures on other banks using regression models and different modern-day approaches.

However, only limited research has been dedicated to understanding and exploring the impact of bank failures on other geographically proximate banks using event studies. The impact of such big failures on the local banking industry and the impact of such news on abnormal return is something that still needs more attention and exploration. Therefore, we aimed at filling this research gap by understanding the impact of bank failure on geographically proximate banks.

Holding onto the Financial Contagion theory proposed by Allen, F., & Gale, D. (2000), we expect that a failure of a bank in one region can cause significant negative returns to banks in the same area. Thus, our main area of research and hypothesis to be tested is:

H_1 : Bank Failures will have a negative impact on the stock returns of geographically proximate banks.

III. Data

This study investigates the impact of local bank failures on the stock performance of other geographically proximate banks. The dataset consists of:

1. ***Bank Failure Events (2015–2025)***: Downloaded from the FDIC Failed Bank List, this dataset includes all 37 U.S. commercial bank failures from 2015 to 2025, with information on the closing date, institution name, city, state, and acquiring institution.
2. ***Active Bank Registry***: Retrieved from WRDS, this dataset includes currently active banks with associated identifiers such as CUSIP, legal names, location (city/state), and institution start dates.

Using the bank failure dates from the FDIC dataset and matching CUSIPs for nearby active banks from WRDS, an input file was created consisting of event dates and corresponding affected bank CUSIPs. The WRDS U.S. Daily Event Study tool was then used to estimate cumulative abnormal returns (CAR) and other event-based statistics. The event study was generated using an expected return estimation window of 100 trading days, with a requirement of at least 70 non-missing return observations to produce estimates of expected returns. To prevent the likelihood of the expected return estimation being affected by the event induced return variance, a gap of 50 trading days between the estimation window and the event window was implemented in the event study. An event window of 10 days before and after the event date was utilized. A total of 97 events were used in the event study. The following output variables were generated:

- ***EVTIME*** – Number of days relative to the event date within the [-10,10] event window.
- ***RET_M*** – Daily mean total return within the [-10,10] event window.
- ***CAR_M*** – Mean cumulative abnormal return over the [-10,10] event window. The cumulative deviation of a bank's return from the expected return estimated using the Fama-French Model.
- ***ABRET_M*** – Daily mean abnormal return within the [-10,10] event window.

- ***ABRET_T & CAR_T*** – Cross sectional t-statistics of abnormal return and cumulative abnormal return to test for statistical significance.

Table 1 below shows the event study output for the variables described above. Based on the t-statistics, multiple days within the event window have statistically significant abnormal returns and cumulative abnormal returns. Where (*) denotes significance at the 10% level, (**) denotes significance at the 5% level, and (***) denotes significance at the 1% level.

<i>Table 1. Event Study Output</i>					
<i>EVTIME</i>	<i>RET_M</i>	<i>CAR_M</i>	<i>ABRET_M</i>	<i>ABRET_T</i>	<i>CAR_T</i>
-10	-0.11%	-0.19%	-0.19%	-1.19	-1.19
-9	0.03%	-0.02%	0.17%	0.65	-0.06
-8	0.32%	-0.07%	-0.05%	-0.27	-0.30
-7	0.03%	-0.26%	-0.19%	-1.05	-1.13
-6	0.45%	-0.07%	0.20%	1.22	-0.26
-5	-0.14%	-0.21%	-0.14%	-0.92	-0.71
-4	0.38%	-0.01%	0.20%	1.66*	-0.02
-3	-0.28%	-0.26%	-0.25%	-1.94*	-0.84
-2	-0.59%	-0.48%	-0.22%	-1.49	-1.33
-1	-0.89%	-0.76%	-0.29%	-1.56	-1.74*
0	-0.77%	-1.29%	-0.52%	-2.33**	-2.44**
1	-0.45%	-2.00%	-0.71%	-2.51**	-3.00***
2	0.55%	-1.60%	0.40%	2.21**	-2.69***
3	0.63%	-1.02%	0.58%	3.00***	-1.57
4	0.39%	-0.81%	0.21%	1.14	-1.24
5	-0.32%	-1.21%	-0.40%	-1.13	-1.93*
6	0.26%	-1.62%	-0.41%	-2.64***	-2.46**
7	-0.31%	-1.65%	-0.03%	-0.22	-2.66***
8	-0.49%	-1.77%	-0.12%	-0.52	-2.64***
9	0.65%	-1.70%	0.07%	0.32	-2.43**
10	0.07%	-1.85%	-0.15%	-0.85	-2.65***

Table 2 below shows summary statistics for abnormal returns computed across the full event window. The sample includes 97 events with at least 70 valid return observations in the estimation window.

<i>Table 2. Summary Statistics</i>				
Event Day	Mean Abnormal Return	Standard Deviation	Min Abnormal Return	Max Abnormal Return
-10	-0.19%	1.57%	-7.10%	3.90%
-9	0.17%	2.60%	-22.07%	5.80%
-8	-0.05%	1.87%	-4.64%	13.83%
-7	-0.19%	1.83%	-5.27%	11.45%
-6	0.20%	1.58%	-2.61%	8.91%
-5	-0.14%	1.49%	-5.34%	5.77%
-4	0.20%	1.18%	-2.04%	4.32%
-3	-0.25%	1.28%	-3.81%	4.51%
-2	-0.22%	1.45%	-4.16%	5.49%
-1	-0.29%	1.80%	-4.79%	5.34%
0	-0.52%	2.20%	-10.28%	4.69%
1	-0.71%	2.80%	-12.91%	7.25%
2	0.40%	1.78%	-3.57%	9.76%
3	0.58%	1.91%	-4.41%	5.70%
4	0.21%	1.82%	-3.72%	9.74%
5	-0.40%	3.50%	-9.19%	30.52%
6	-0.41%	1.52%	-7.35%	4.14%
7	-0.03%	1.45%	-3.29%	5.89%
8	-0.12%	2.22%	-7.59%	11.51%
9	0.07%	2.20%	-5.79%	12.91%
10	-0.15%	1.77%	-9.44%	4.20%

The figure below displays the mean cumulative abnormal returns (CAR) across the [-10, 10] event window with 95% confidence bounds. The sharp drop following Day 0 highlights a statistically significant negative return, suggesting market reaction to nearby bank failures.

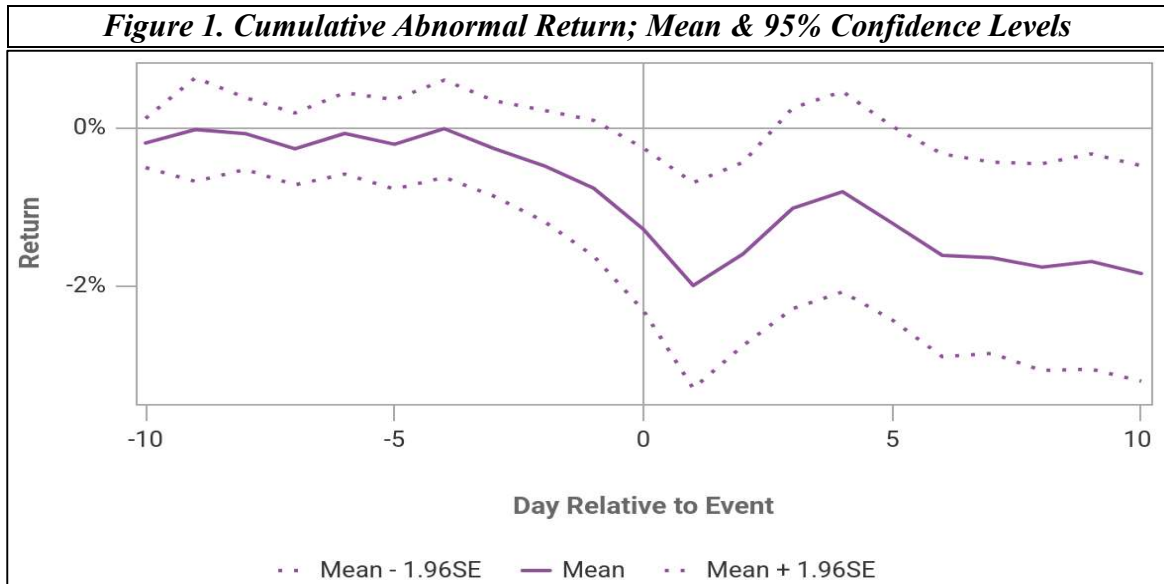


Table 3 shows the annual distribution of failures from 2015 to 2025. Failures were concentrated in the earlier years of the sample, particularly in 2015 and 2017, which each saw 8 bank failures. There were notably fewer failures during 2018 and 2019, aligning with broader macroeconomic stability in those years. The latter part of the sample, post-2020, shows a moderate resurgence in failures, possibly reflecting residual economic stress from the COVID-19 pandemic and rising interest rate environments.

Table 3. Time Distribution of Bank Failures		
Year	Bank Failures	Percent of Total
2015	8	22%
2016	5	14%
2017	8	22%
2018	0	0%
2019	4	11%
2020	4	11%
2021	0	0%
2022	0	0%
2023	5	14%
2024	2	5%
2025	1	3%

Table 4 displays the geographic breakdown of bank failures across U.S. Census regions. The Southeast and Midwest regions dominate the distribution, accounting for the majority of bank failures. These regions tend to feature a higher density of small to mid-sized community banks, which may be more vulnerable to localized shocks, poor loan performance, or concentrated economic exposures. In contrast, the Northeast and West regions show relatively fewer bank failures during this period, possibly reflecting stronger urban diversification, regulatory oversight, or institutional resilience.

This regional concentration supports the underlying hypothesis of the study: that financial contagion from bank failures is often geographically bounded. The clustering of failures in specific regions increases the likelihood that neighboring institutions will be viewed by markets as similarly exposed, triggering negative abnormal returns. This pattern reinforces the importance of monitoring regional banking health, not just individual institutions, when evaluating financial stability and contagion risk.

<i>Table 4. Geographic Distribution of Bank Failures</i>		
<i>U.S. Census Region</i>	<i>Bank Failures</i>	<i>Percent of Total</i>
Midwest	14	38%
Northeast	5	14%
South	13	35%
West	5	14%

The event study sample suffers from a few data limitations. To isolate the sample from market wide disruptions caused by economic downturns and recessions that can affect bank stock returns uniformly, this research focused on bank failures in the previous 10 years from 2015-2025. The U.S. saw only 37 commercial bank failures in this period providing a relatively small sample size to conduct the event study. The nature of the event study also limited what institutions could be

analyzed. By only including publicly traded banks with a usable CUSIP, private or unlisted banks without the necessary data on stock returns are excluded from the event study sample even if they could have also been affected by the nearby bank failure. Spillover effects on neighboring banks are only measured in terms of short-run abnormal returns, ignoring the long-term effects of the bank failures.

IV. Methodology

To estimate the effect of a bank failure on the stock returns of peer banks in the same geographic area, an event study methodology was employed. This approach quantifies the abnormal return of the affected banks returns beyond what would be expected based on market and risk factor exposures.

The Fama-French Three Factor Model was used as the benchmark return model to control for market-wide movements, size effects, and value effects. This model is suitable given its established empirical power in explaining cross-sectional return variation.

The mathematical framework of the Fama-French Model used to calculate the expected return is modeled as:

$$E(R_{it}) = \alpha_i + \beta_{iM}R_{mt} + \beta_{iSMB}SMB_t + \beta_{iHML}HML_t + \varepsilon_{it} \quad (1)$$

Where R_{it} is the return on bank i on day t , R_{mt} is the return on the market portfolio, SMB_t is the return on the small-minus-big size factor, and HML_t is the return on the high-minus-low book-to-market factor.

The abnormal return is calculated as the difference between the actual and expected return:

$$AR_{it} = R_{it} - E(R_{it}) \quad (2)$$

The cumulative abnormal return (CAR) is calculated as the summation of abnormal returns:

$$CAR_i = \sum_{t=\tau_1}^{\tau_2} AR_{it} \quad (3)$$

Where $\tau_1 = 10$ and $\tau_2 = 10$ using a $[-150, 50]$ expected return estimation window with a minimum of 70 return observations required.

The statistical significance of the CARs was assessed using cross sectional t-statistics which help validate whether observed abnormal returns are likely due to the event rather than random fluctuations.

V. Empirical Results

This section presents the results of our empirical analysis of the effects of bank closures on stock performance of banks in the proximity of the closed banks. We seek to analyze abnormal return behavior following the announcement of 37 bank closures from 2015 to 2025. The goal is to see whether the closure of these banks has a statically significant impact on the stock price of the nearby commercial banks, and whether it will increase, or decrease the stock price. We also look to find how far in advance of the event day stock prices are affected.

The approach for this analysis is a simple event study. Event studies are commonly used in finance to examine the effect of significant events on the value of a particular security, in our case stock price, to see how it can react to impactful events. The event in question, and what we are studying, is the announcement of a commercial bank closure or failure. The announcement of the bank closure is marked as 0 in our event study.

The abnormal return is calculated as the actual return on a bank's stock minus the expected return, predicted using the Fama-French Three Factor model. The model was estimated over a 100-day

window prior to the event and separated by a 50-day gap. In the event study, we use a 21-day period to determine how the closure affects a proximity bank's stock price, 10 days before and after the event, including the event date. Below are the results of our event study analysis.

Looking at Figure 1 showing the cumulative abnormal return, we can see there is a steady state of return until about 4 days before the event day, where it starts to decline. Immediately after the event day and about a day after, the cumulative abnormal returns drop even more to their lowest point, followed by a slight increase in price to a peak around day 4, and then followed by a steady drop in the cumulative abnormal returns.

What this shows us is investors interpret the failure of the bank as a signal towards broader regional instability, which in turn affected the perceived risk of neighboring banks. In analyzing the decline in cumulative abnormal returns before the event day we can see investors are less confident in neighboring banks which starts the decline in cumulative abnormal returns. This could be due to insider trading or rumors. As rumors spread of a bank failure, or even insider knowledge that a bank will fail, investors will start to sell their securities in neighboring banks in anticipation of the failure. Astute financial analysts could also see the impending failure by looking at financial reports of banks and seeing that they are troubled and can likely to fail.

After the event day we can see the price continuing to fall, but the event day doesn't show necessarily a drastic decrease, indicating the cumulative abnormal returns we already falling at a steady rate before the event day. They soon begin to rise which can be due to an overreaction bias towards the news. Since banks are highly regulated and all deposits are insured in most cases, the federal government has put in place several security measures to prevent bank runs and other banks from failing due to another bank failure. This could be a reason why the cumulative abnormal returns began to rise after a sharp selloff, as investors realize this. What is notable however, is that

after the rise in cumulative abnormal returns, they begin to drop steadily even more to about the same level they were on the event day.

For shareholders of nearby banks, these market reactions can translate into substantial losses, particularly for smaller banks with less trading volumes and limited diversification. These findings are important for regulators as this evidence implies that a single bank failure can reverberate past its own failings, introducing short-term instability to other banks in the region of failure. For investors, these results highlight the need to further diversify their portfolio across regions, so that one bank failure doesn't affect their entire holdings.

Limitations and Caveats

While this study provides empirical evidence of spillover effects from bank failures to geographically proximate commercial banks, several limitations must be acknowledged. It looks at only 37 bank failures between 2015 and 2025, a relatively calm period for the U.S. banking sector. While this helps reduce interference from large economic shocks, the small sample size limits how much we can generalize the findings. Second, the analysis includes only publicly traded banks because of data availability, leaving out private and smaller institutions that might also be affected by nearby failures. This could mean the actual impact is even larger than what we observed. Third, the event window only covers short-term effects—10 days before and after the failure, so it doesn't capture longer-term outcomes like changes in lending or customer behavior. Fourth, the study assumes that markets react quickly and rationally to news, but real-world investor behavior can be influenced by emotions or delayed responses. Lastly, the analysis defines proximity based on physical location, without considering deeper connections like shared customers or financial ties, which may play a bigger role in how risks spread between banks.

VI. Conclusion

Summary of findings

This study examined whether commercial bank failures have a measurable effect on the stock performance of nearby banks. By analyzing 37 bank failure announcements from 2015 to 2025 using an event study methodology based on the Fama-French Three Factor model, we found clear evidence of statistically significant negative abnormal returns for geographically proximate banks. These effects began a few days before the public announcements, likely due to market speculation or insider information and intensified immediately afterward. Cumulative Abnormal Returns (CAR) reached their lowest point within two days of the failure, indicating a swift market reaction. Although there was a brief rebound, likely driven by investor confidence in federal safety nets, CARs remained depressed in the following days, suggesting lingering concerns about regional stability and the potential for further contagion. These findings emphasize the broader implications of localized bank failures and the importance of monitoring spillover effects in financial markets.

Implications

This study gives real-world evidence supporting the Financial Contagion Theory by showing how the failure of one bank can affect nearby banks, even when the overall economy is stable. It also highlights how markets often move even before any official news comes out. This kind of behavior fits with ideas from behavioral finance, where investors might jump to conclusions or overreact when they sense something's wrong, even if it hasn't been confirmed yet. The study's results also reveal that regionally focused and smaller banks are particularly exposed to spillover risks stemming from neighboring bank failures. This highlights the importance of not only monitoring their own financial health but also staying aware of vulnerabilities within their local financial

ecosystem. Bank leadership should take steps to strengthen communication with stakeholders, reinforce internal controls, and build buffers that can help sustain confidence during times of localized disruption. Likewise, the regulators should not only focus on individual banks but also consider the broader network effects to maintain stability. They should improve communication during bank closures, strengthen efforts to reassure depositors, and closely monitor how banks within the same region are connected.

Future Research

This study opens several avenues for future research. First, scholars could investigate the long-term consequences of geographically proximate bank failures, particularly their effects on lending practices, capital costs, and interbank trust dynamics. Second, future work could apply network analysis techniques to uncover spillover pathways that extend beyond physical proximity, such as shared funding arrangements or interbank exposures. Third, there is a need to examine these phenomena in international contexts, where variations in regulatory frameworks, market structures, and investor behavior may yield different contagion patterns.

In summary, although the incidence of bank failures has been relatively low in the post-crisis period, their regional ripple effects remain significant. A deeper understanding of these localized dynamics is essential for designing policies that enhance financial stability and mitigate systemic risk.

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